



High-performance Computing on a 3×3-Inch Board

The Beagle Board is a generic board powered by an OMAP3530 processor. It is backed by a highly-active community, which maintains its own portal that invites discussion, queries, ideas and other inputs from all over the world—the first of its kind.

The Beagle Board, an open source initiative by Texas Instruments, was launched to the public around a year back. It was developed by a small team at Texas Instruments, led by Khasim Syed Mohammed, who's the lead developer for open platforms at TI. It's called the Beagle Board because of its dimensions. A beagle is a breed of short and cute dogs, which the team found similar to this 3x3 inch (7.6x7.6 cm) board. It's a low-cost, fan-less, single board computer with a high performance embedded platform, which is based on Texas Instruments' OMAP3530 technology.

"The idea of a Beagle Board was born a year back and I have been involved in this initiative from the very beginning. It took six months for the team to convert this idea into a reality. I started with hardware enablement for this initiative with boot loaders, the Linux kernel and drivers, validation, documentation, promotions, etc," notes Syed. "Ever since its launch, I've been promoting Beagle Board among university students and the small-scale IT sector in India, continuing in enabling a global community to be built around OMAP3 technology. Apart from other activities, I've also

been working to demonstrate the capability of OMAP3 architecture in the medical, infotainment, portable navigation and low-power industrial sectors."

The interesting point to note here is that the board is targeted at the open source developers, which seemed obvious considering the lead developer's passion for open source. "I was interested in computers right from my early days. After my school, I started learning computers and during my internship on TI's DSP, my curiosity for embedded applications increased enormously. Moving forward, I found open source to be more useful since it was free and gave me all the necessary software blocks 'with source' to build end-to-end solutions," reveals Syed. "I started as a user/consumer of open source software and forums. While working closely with the OMAP Linux community, I found significant gaps in our offerings to the open community. I joined Jason Kridner and Gerald Coley at TI to bridge and address these gaps through beagleboard.org."

He adds: "I am an enabler and a contributor of open software in the community. The demand for consumer devices is growing